# MUSTAFA SABİT EROL

Computer Engineer
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## SUMMARY

Computer Engineering graduate from KTO Karatay University (GPA 3.41/4.00, Honor Student). Internship and project experience in visual SLAM and camera calibration with ROS 2, ESP32-based embedded systems, real-time network programming and metaheuristic optimization (ABC, ACO). Seeking a role in embedded software, C++ software development or image processing in the defense industry.

## TECHNICAL SKILLS

**Programming Languages:** C, C++ (Arduino / ESP32), C#, Python, Java, SQL
**Robotics and Vision:** ROS 2, ORB-SLAM2, RTAB-Map, Webots, LiDAR, camera calibration, point clouds
**AI and Algorithms:** Metaheuristic optimization (ABC, ACO), A* pathfinding, context steering
**Embedded Systems:** ESP32, ESP8266, Arduino, MQTT, infrared communication, Wi-Fi
**Tools and Platforms:** Linux (Ubuntu), Git, Unity, Photon Fusion, socket programming, multithreading, Gazebo and MATLAB (basic)

## EDUCATION

**KTO Karatay University — B.Sc. in Computer Engineering** 2022 – June 2026
- Konya, Türkiye | GPA: 3.41 / 4.00 — Honor Student

## EXPERIENCE

**Intern — SIMSOFT Computer Technologies** July 2025
*Ankara, Türkiye*
- Set up a ROS 2 environment on Ubuntu and worked with the team on mapping and localization using dual-camera visual SLAM.
- Calibrated and synchronized two cameras using a checkerboard pattern; generated 3D point cloud maps.
- Analyzed the ORB-SLAM2 and RTAB-Map pipelines (feature extraction, pose estimation, loop closure, pose graph optimization) at ROS node and topic level and prepared technical documentation for the team.

**Summer Intern — AIMSOFT Software** July 2024
*Niğde, Türkiye*
- Implemented Artificial Bee Colony (ABC) and Ant Colony (ACO) algorithms in Python for assembly line optimization and compared them on simulated assembly lines by solution quality and convergence.

## PROJECTS

**DuelKeep — Online Mobile PvP Game (Graduation Project, Solo)** 2025 – 2026
*Unity 6 (URP 2D), C#, Photon Fusion 2, Unity Gaming Services*
- Developed and released a real-time 1v1 online action game on Google Play (TR/EN); built server-authoritative networking with Photon Fusion 2, version-separated matchmaking and a skill-based league system.
- Built the backend on Unity Gaming Services (Authentication, Cloud Save, Leaderboards, Cloud Code) with server-side sanity checks on league points.
- Built an adb-based on-device profiling harness (frame times, profiler captures, temperature) and removed bottlenecks, taking Raid mode from 17 to a stable 60 fps.
- Designed a unified isometric depth-sorting model that cut sorting errors from 9.8% to 0.6%; wrote automated collider-audit tooling and added 369 wall bridges.
- Designed bot AI with A* pathfinding and context steering; wrote 112 automated unit tests.

**BrickRift — Mobile Brick-Breaker Game (Personal Project)**
*Unity 6, C#, Android*
- Built an offline Android brick-breaker with table-tennis paddle physics (spin and smash), 100 levels across 10 worlds and TR/EN localization; wrote tools for level/skin generation and physics testing.

**Autonomous Vehicle Project — Simulation and Testing (Team Project)**
*Webots, ROS 2, LiDAR*
- Set up the autonomous vehicle simulation in Webots and tested its motion and environment perception behaviour.
- Visualized LiDAR data and validated obstacle detection and vehicle responses in simulation before physical prototype testing.

**Embedded and IoT Projects**
*ESP32, ESP8266, MQTT, C++ (Arduino)*
- Built a system that records remote control signals with an infrared receiver, replays them through an ESP32-driven IR LED and is controlled from a local web interface.
- Built an automatic pet feeder based on ESP8266, controlled remotely over MQTT.

## COMPETITIONS AND CERTIFICATES

**TEKNOFEST UAV Competition — Team Bangu, "Canhıraş" Project (Avrasya University)** 2022 – 2023
- Passed the Critical Design Review (CDR) stage as a team.

**LEVEL UP AI | ROKETSAN AI Hackathon — AI Workshop (3 days, certificate of participation), Coderspace** August 2026

## LANGUAGES AND ADDITIONAL

**Languages:** Turkish (native), English (advanced; YÖKDİL 86.25, exempted from university English prep) | Military Service: Deferred | Driving License: A2, B